

Actividad 1: Información de Países (REST Countries API)

Objetivo

Introducir el concepto de REST API, practicar peticiones GET, interpretar JSON y consumir una API pública con Postman y Python.

API gratuita (CON API Key)

Fase 1 – Exploración de la documentación

- Base URL: `https://restcountries.com/`
- Regístrate y obtener una API KEY

REST Countries

APIs Docs Log in Get an API key

Get an API key
Create a free account to access your API key.

FULL NAME
Gabriel Barrón Rodríguez

EMAIL
gbarron@utng.edu.mx

PASSWORD
.....

CONFIRM PASSWORD
.....

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¡Operación exitosa!

Cloudflare

Create account

https://restcountries.com/sign-up#

- Guarda tu API Key de tal manera que no se te olvide
- Abre tu consola y teclea el siguiente comando

curl "https://api.restcountries.com/countries/v5?q=canada&pretty=1" -H "Authorization: XXXXXX"

Get an API key

Create a free account to access your API key.

FULL NAME

Jairo Haziél Rosas Enríquez

EMAIL

1225100376.jhre@gmail.com

PASSWORD ⓘ

••••••••

CONFIRM PASSWORD

••••••••

☒ I agree to the [Terms of Service](#) and [Privacy Policy](#).

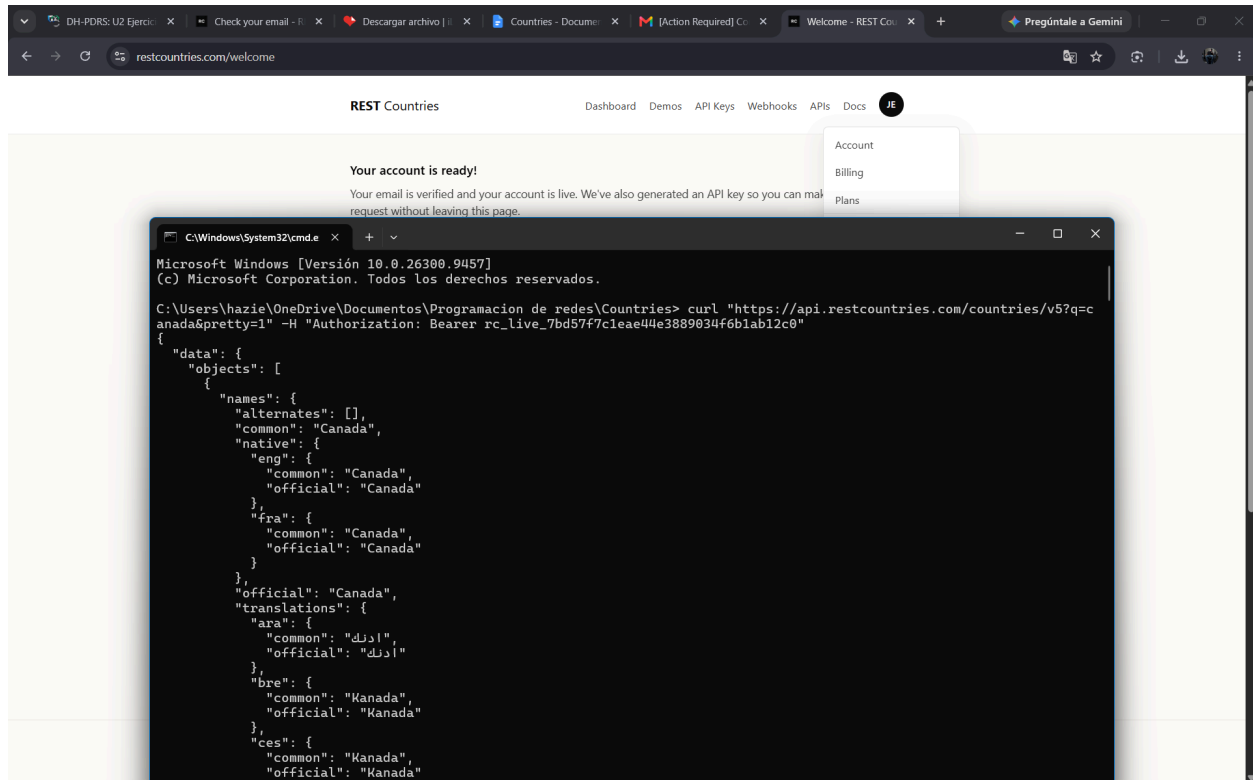


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Privacidad • Ayuda

Create account

- Pega tu resultado de manera que se visualice tu perfil



Postman

- Abre postman
- Crea una colección de API REST
- Agrega una nueva petición
- Selecciona el método **GET** y localiza la sección de HEADERS, agrega la clave **Authorization** y pega su contenido "Bearer API_KEY"

Introducción a REST API

The screenshot displays the Postman application interface. The top bar shows the workspace name "Jairo Haziel Rosas Enriquez's Workspace" and a search bar. The left sidebar contains a "COLLECTIONS" panel with "Países" expanded, showing "Canada" selected. Below it are "Programacion Redes" and "Unidad2_Act2_Jairo". The "ENVIRONMENTS" panel is also visible with options like "SPECS", "MOCKS", "DATASETS", "TOOLSETS", "DOCUMENTS", and "FLOWS".

The main workspace shows a REST client request configured as follows:

- Method:** GET
- URL:** `https://api.restcountries.com/countries/v5?q=canada&pretty=1`
- Headers:** 7 headers, with "Authorization" set to `((bearer_token_17zf))`.
- Body:** The response is displayed in JSON format, showing a 200 OK status with a response time of 697 ms and a size of 3.26 KB. The JSON structure is as follows:

```
1 {
2   "data": {
3     "objects": [
4       {
5         "names": {
6           "alternates": [],
7           "common": "Canada",
8           "native": {
9             "eng": {
10              "common": "Canada",
11              "official": "Canada"
12            },
13            "fre": {
14              "common": "Canada",
15              "official": "Canada"
16            }
17          },
18          "official": "Canada",
```

The bottom status bar indicates "Ready to go" and provides links to "Globals", "Vault", and "Tools".

Código Python

```
import requests

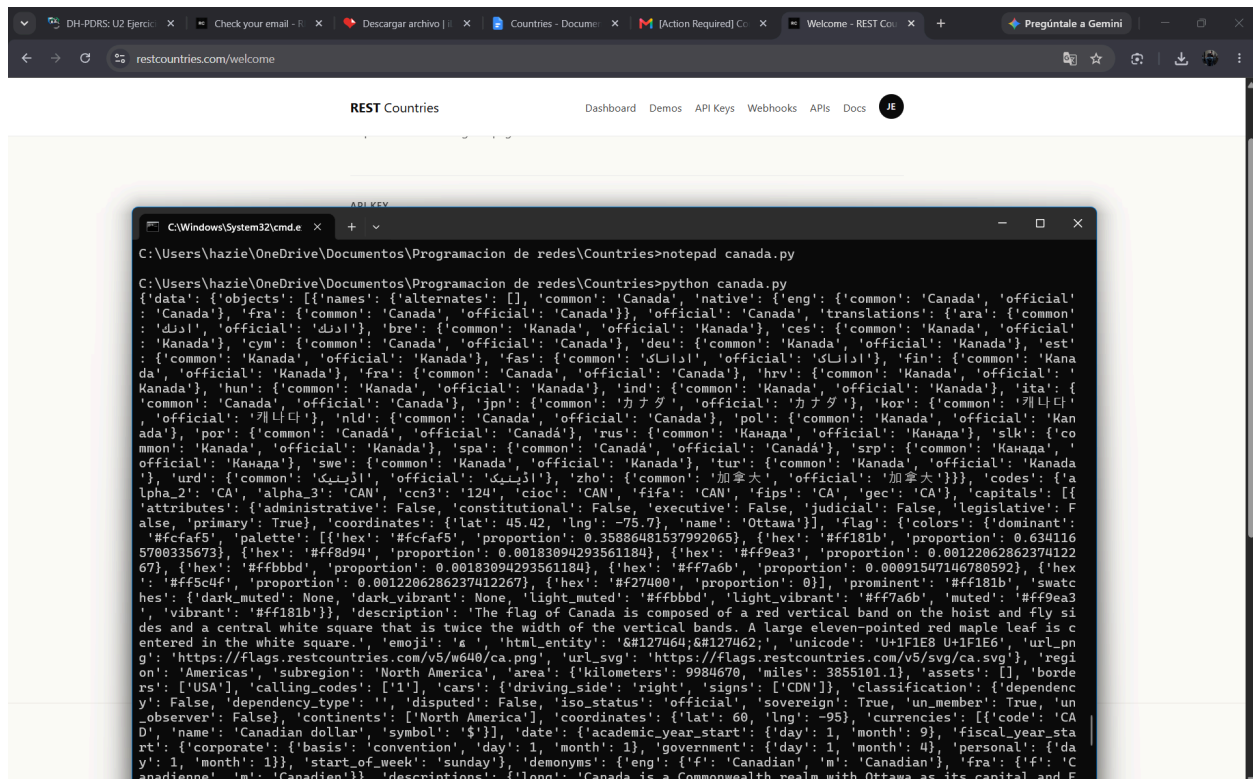
params = {
    "q": "canada",
    "pretty": "1"
}

url = 'https://api.restcountries.com/countries/v5'

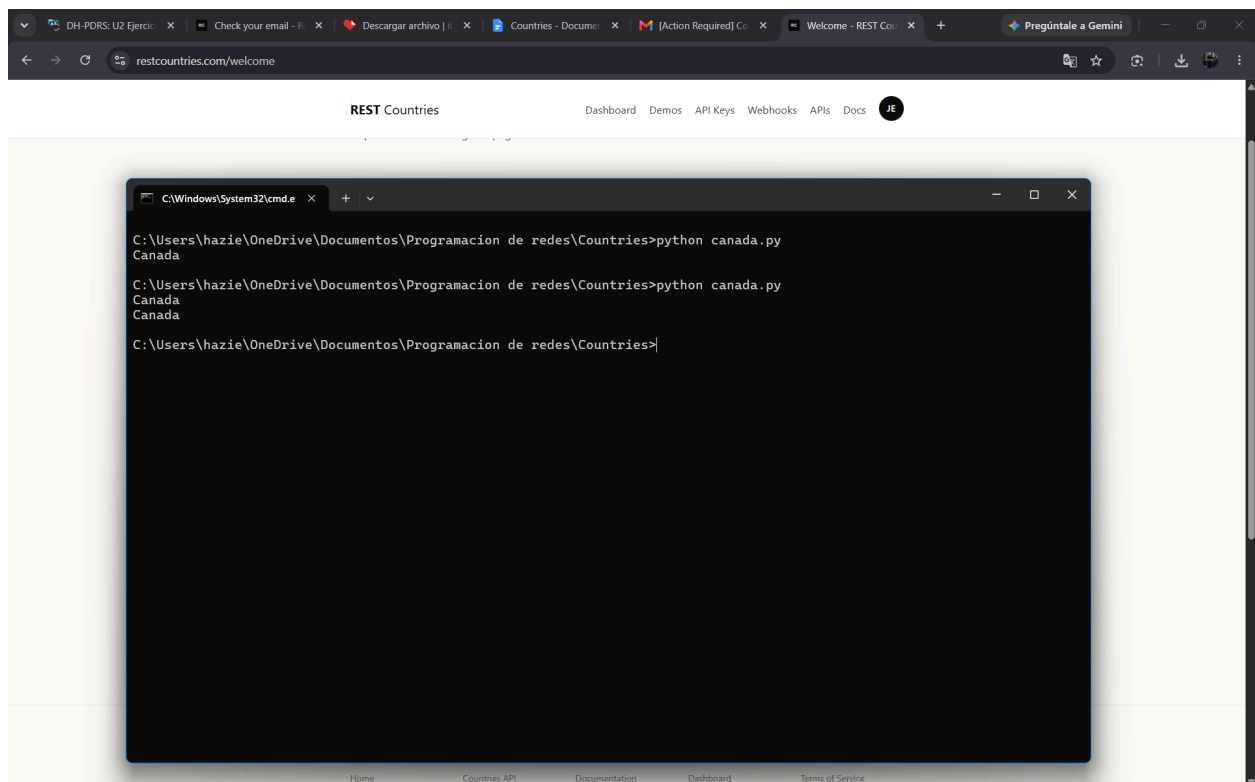
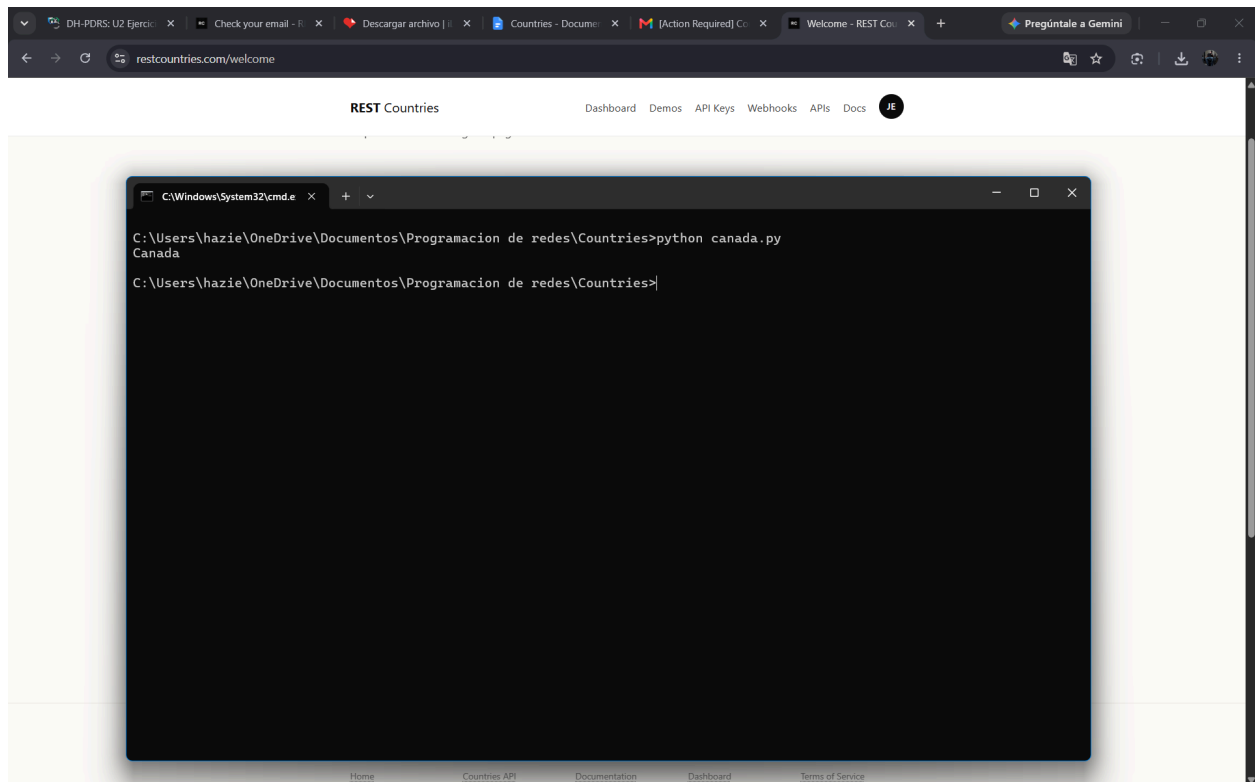
headers = {

    "Authorization": 'Bearer
rc_live_5c958707a45848188d0e68bea868d00d'
}

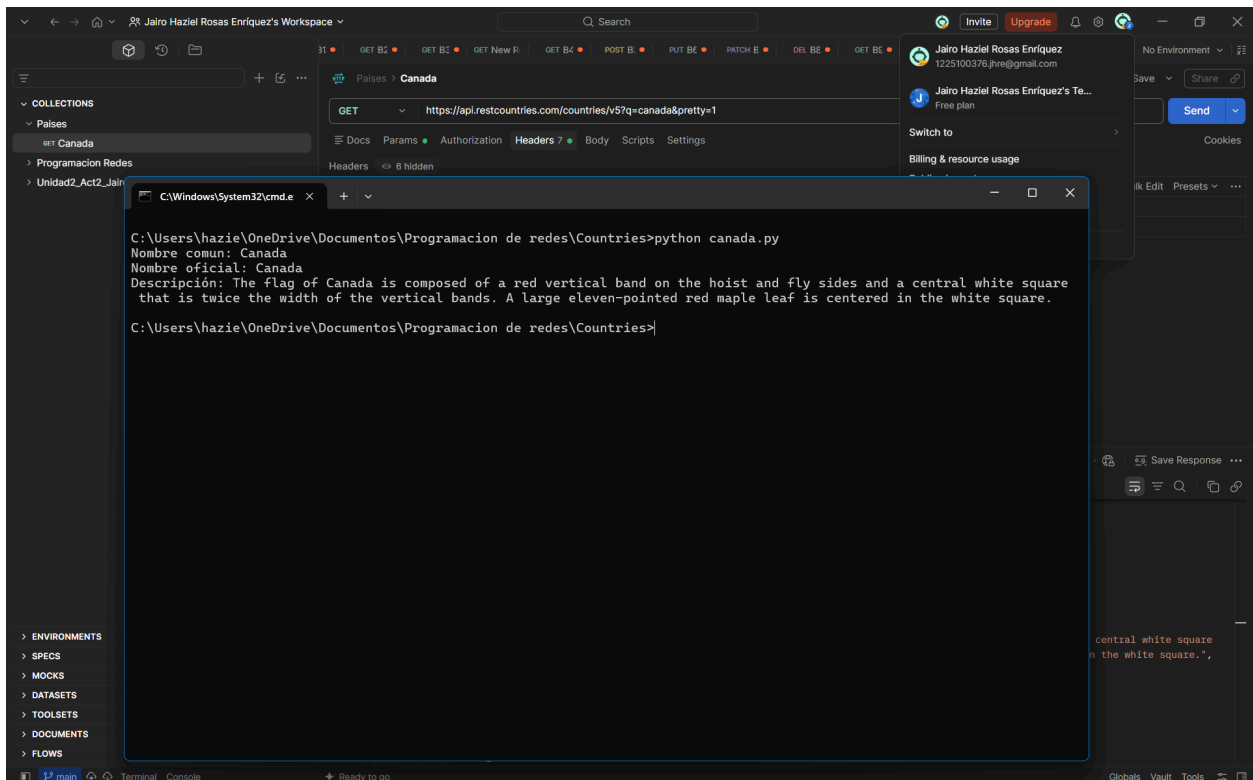
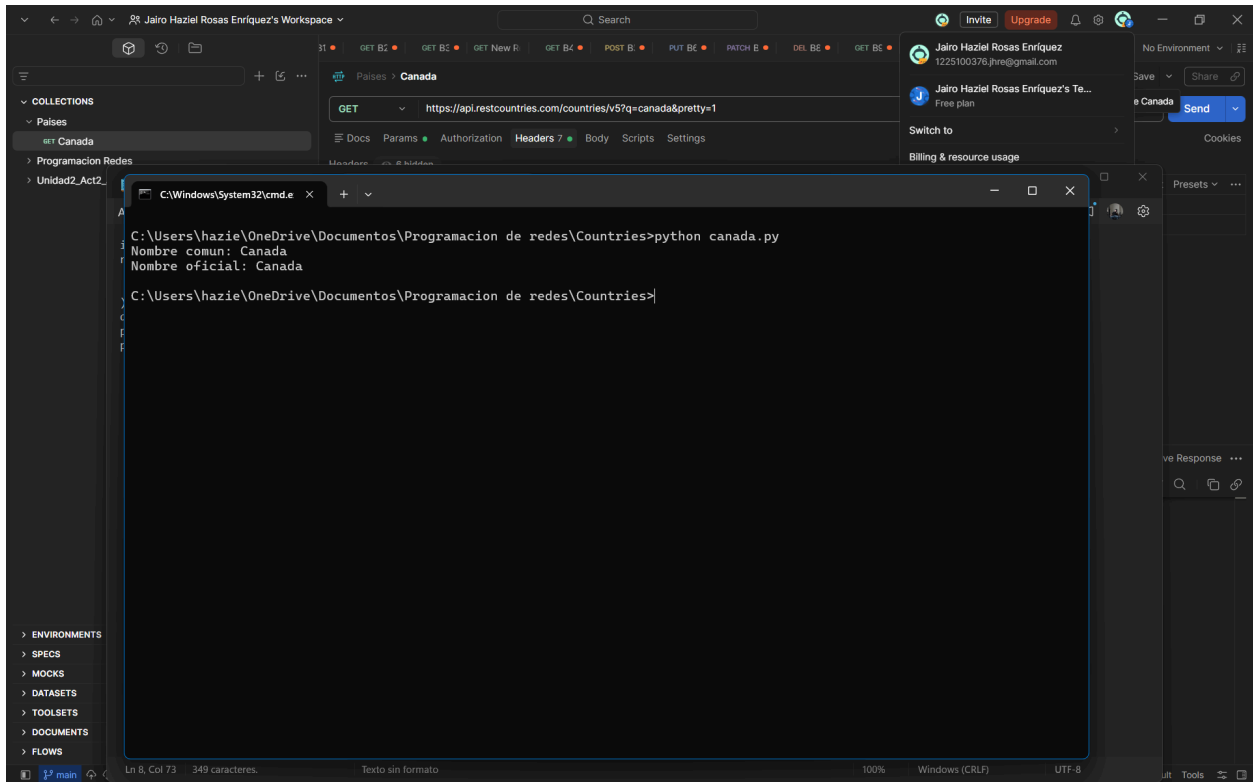
response= requests.get(url, params=params, headers=headers)
if response.status_code == 200:
    data = response.json()
    print(data)
else:
    print(f"Error {response.status_code}: {response.text}" )
```

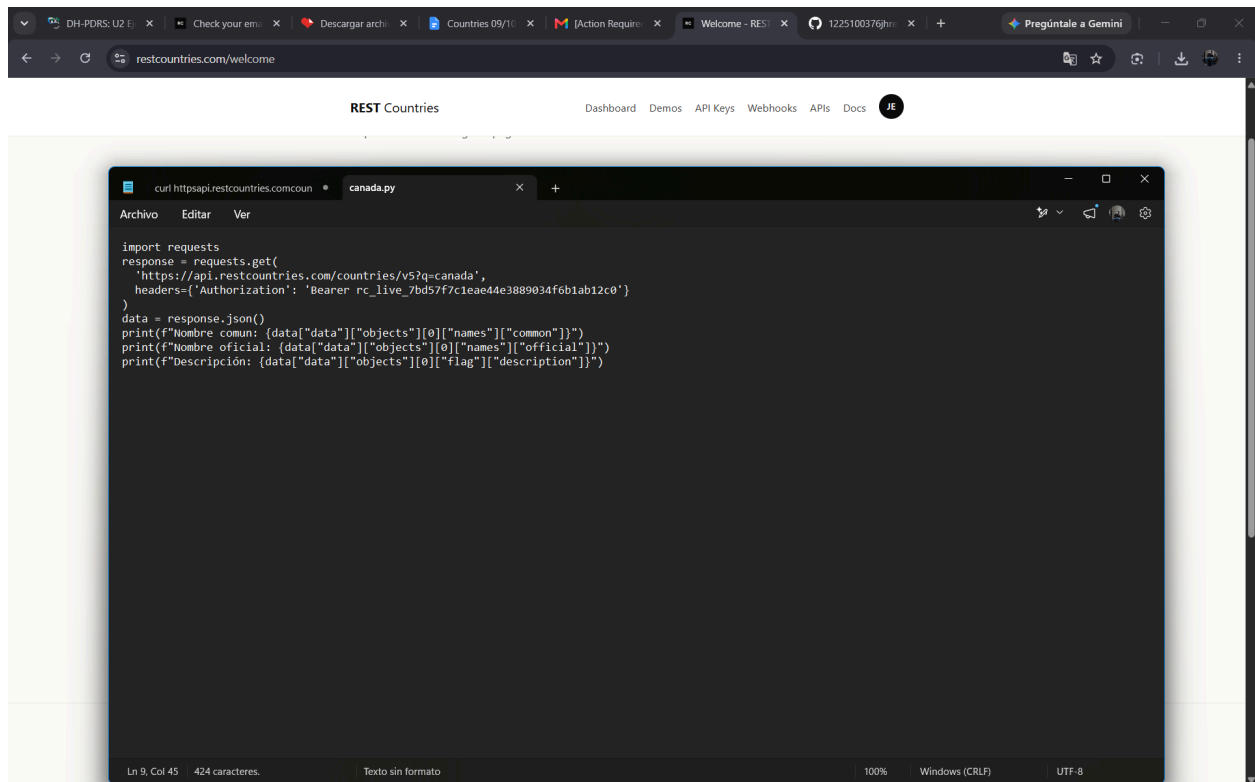


Introducción a REST API



Introducción a REST API





Fase 2 – Desafíos (4-6)

Desafío 1 (curl, Postman y python)

- Realiza una petición CURL para México
- Realiza una petición Postman
- Imprime su nombre común, nombre oficial, nombre de la capital, región, continente, descripción y economía

```
C:\Users\Usuario\Downloads\python>python countries.py
Nombre Común: Mexico
Nombre Oficial Mexico
```

Ejemplo para acceder a los datos

```
print(f"Nombre Común: {data['data']['objects'][0]['names']['common']}")
```

```
print(f"Nombre Oficial {data['data']['objects'][0]['names']['common']}")
```